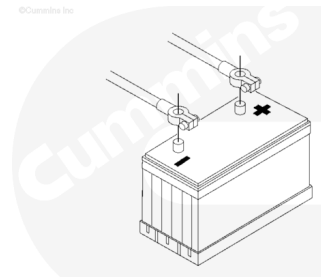


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



13800050

Disconnect the batteries.

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



ck800wa

- Remove the transmission. Refer to the original equipment manufacturer's instructions
- Remove the flywheel or flexplate. Refer to Refer to Procedure Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html>) or Refer to Procedure Procedure 016-004 (</qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html>)
- Remove the rear crankshaft seal. Refer to Refer to Procedure Procedure 001-024 (</qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html>)

⚠ WARNING ⚠

Support the rear of the engine using the rear support attached to the rear of the cylinder block. Failure to support the engine can cause personal injury.

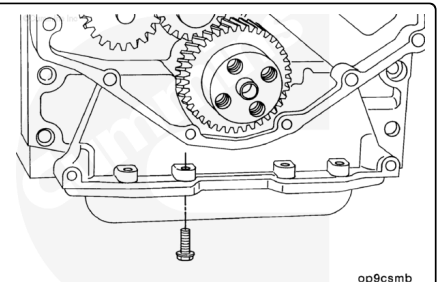
- Remove the flywheel housing. Refer to Refer to Procedure Procedure 016-006
(/qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html)
- Remove the fuel transfer pump. Refer to Refer to Procedure Procedure 005-045
(/qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html)
- Remove the fuel injection pump. For rotary fuel injection pumps, refer to Refer to Procedure Procedure 005-014
(/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html). For inline fuel injection pumps, refer to Refer to Procedure Procedure 005-012
(/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html).
- If equipped, remove the hydraulic pump unit. Refer to Refer to Procedure Procedure 009-016
(/qs3/pubsys2/xml/en/procedures/40/40-009-016.html).

Service Tip: Engines equipped with air compressors may require that the air compressor be timed to the engine. To ensure that the air compressor is properly timed when the camshaft gear is later installed, scribe an alignment line on the air compressor and camshaft gear before removing the camshaft gear or air compressor.

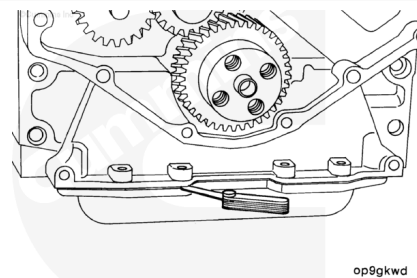
- If equipped, remove the air compressor. Refer to Refer to Procedure Procedure 012-014
(/qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html)
- Remove the camshaft gear and camshaft. Refer to Refer to Procedure Procedure 001-008
(/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html)

Remove

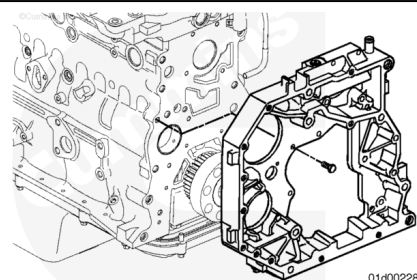
Remove the four (4) oil pan to rear gear housing capscrews.



Using a feeler gauge, separate the lubricating oil pan gasket from the gear housing.



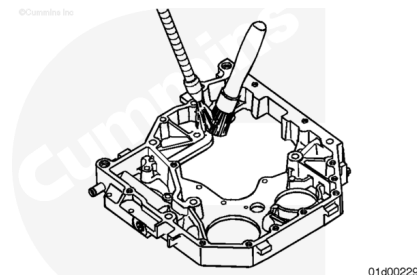
Remove the rear gear housing capscrews and housing.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

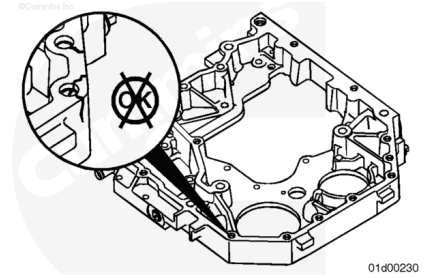
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the rear gear housing with solvent. Dry with compressed air.

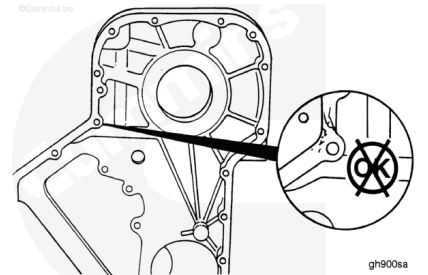
Clean the oil supply hole for the accessory drive.

Note : The rear gear housing has oil drain and supply passages designed into the housing. Make sure these passages are clean and free of debris.

Inspect the rear gear housing for signs of leakage or any other damage.

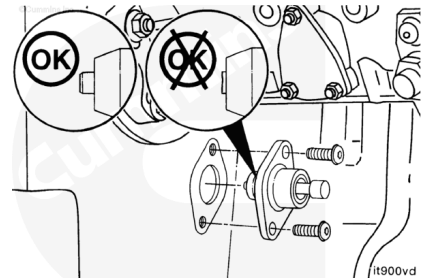


Note : If the rear gear housing is cracked at the accessory drive hole or fuel injection pump mounting location, double check the mounting brackets, fasteners and component installation practices.

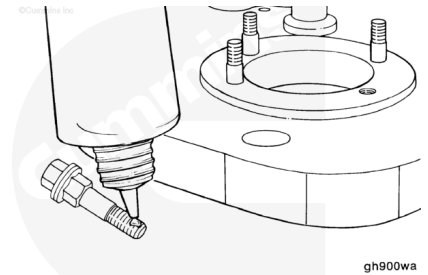


Inspect the timing pin housing and pin for damage.

Note : Do not remove the timing pin assembly unless it is damaged or leaking, or the gear housing is being replaced. Refer to Procedure 001-049 (</qs3/pubsys2/xml/en/procedures/40/40-001-049.html>).



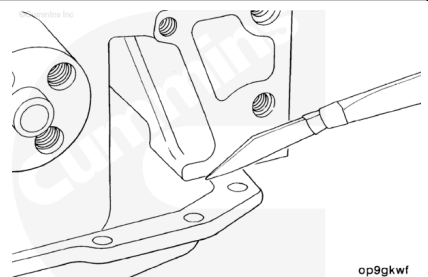
If the fuel injection pump studs are damaged or being installed in a new housing, coat the threads with Loctite 242. Use two nuts locked together to remove and install the studs.



Install

For engines equipped with a paper gasket style oil pan, inspect the oil pan gasket for damage.

Note : If the engine is equipped with a suspended oil pan with a rubber oil pan gasket and the gasket is found to be damaged, the oil pan must be removed and the entire oil pan gasket replaced. Refer to Procedure 007-025

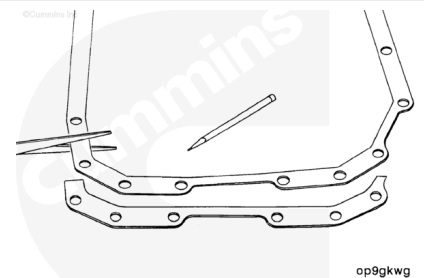


(/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html). If the engine used sealant only between the gear housing and the oil pan, remove the old sealant and apply new sealant, part number 3164070, to the oil pan mounting flange.

If the pan gasket is torn, it can be repaired.

Cut the torn gasket off even with the front of the cylinder block.

Using the old gasket as a pattern, cut the front section of a new gasket to the same size.

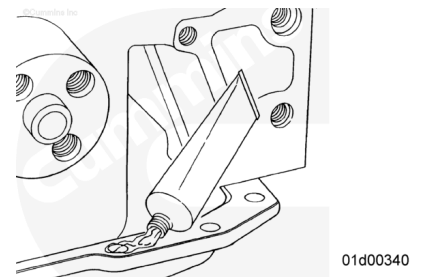


Note : The gear housing must be installed within 10 minutes of applying the sealant.

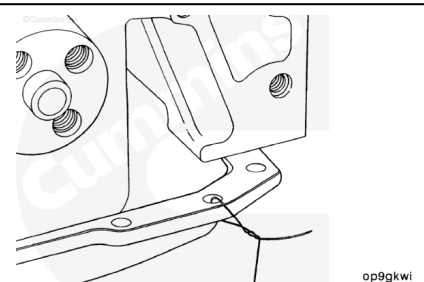
Clean the sealing surfaces.

Coat the new gasket on both sides with sealant, Part Number, 3164067.

Be sure there is a bead of sealant at the intersecting joint of the cylinder block, oil pan, and gear housing.

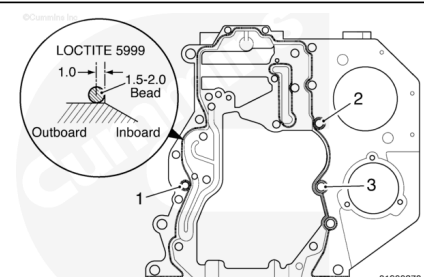


Use common thread or a very fine wire to hold the new gasket splice in position as illustrated.



⚠ CAUTION ⚠

Make sure to only apply sealant to the areas specified. Failure to do so may block oil passages to the accessory drive and/or cause a loss of oil pressure. This will result in severe engine damage.



Apply a 1.5 to 2.0 mm [0.06 to 0.08 in] wide bead of sealant, Part Number 3164070, to the block side of the gear housing in the path illustrated and install the rear gear housing capscrews and housing.

Note : Make sure to apply a bead of sealant around the mounting capscrew holes shown, locations 1, 2 and 3.

Be sure there is a bead of sealant at the intersection joint of the cylinder block, oil pan, and gear housing.

Note : Install the gear housing within 10 minutes of applying sealant or the sealant will **not** seal correctly. Once installed, allow the to dry for 30 minutes before running the engine.

Tighten the capscrews as shown.

Torque Value:

M12

1. 50 n•m [37 ft-lb]

Torque Value:

M10

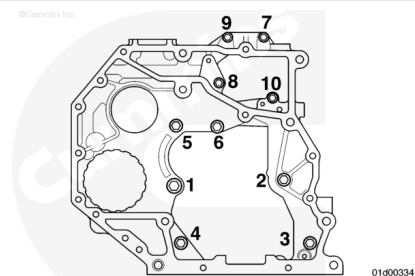
1. 47 n•m [35 ft-lb]

Torque Value:

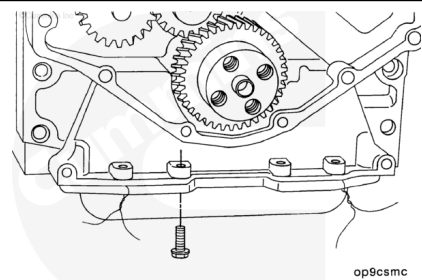
M8

1. 24 n•m [212 in-lb]

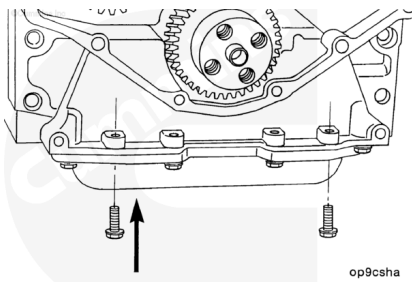
Note : If a rear gear housing and/or timing pin assembly is installed, the timing pin must be accurately located. Refer to Refer to Procedure Procedure 001-049 (/qs3/pubsys2/xml/en/procedures/40/40-001-049.html)



Start the oil pan capscrews in the holes **not** being used to tie the gasket in place.

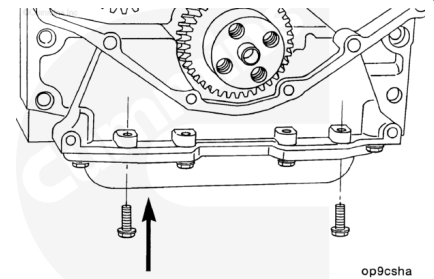


Remove the thread or wire holding the gasket in place.



Install the remaining two (2) oil pan cap screws and tighten.

Torque Value: 28 n·m [21 ft-lb]



Finishing Steps

- Install the camshaft gear and camshaft. Refer to Refer to Procedure Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html>).

Service Tip: Engines equipped with air compressors may require that the air compressor be timed to the engine. To ensure that the air compressor is properly timed when the camshaft gear is installed, a line should have been scribed during when removed. Align this line when installing the air compressor. If the line was **not** scribed or is no longer available, refer to Refer to Procedure Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html>).



- If equipped, install the air compressor. Refer to Refer to Procedure Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html>)
- If equipped, install the hydraulic pump unit. Refer to Refer to Procedure Procedure 009-016 (</qs3/pubsys2/xml/en/procedures/40/40-009-016.html>)
- Install the fuel transfer pump. Refer to Refer to Procedure Procedure 005-045 (</qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html>)

- Install the fuel injection pump. For rotary fuel injection pumps, refer to Refer to Procedure Procedure 005-014 (/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html). For inline fuel injection pumps, refer to Refer to Procedure Procedure 005-012 (/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html).

⚠ WARNING ⚠

Support the rear of the engine using the rear support attached to the rear of the cylinder block. Failure to support the engine can cause personal injury.

- Install the flywheel housing. Refer to Refer to Procedure Procedure 016-006 (/qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html)
- Install the rear crankshaft seal. Refer to Refer to Procedure Procedure 001-024 (/qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html).

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Install the flywheel or flexplate. Refer to Refer to Procedure Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html) or Refer to Procedure Procedure 016-004 (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html).

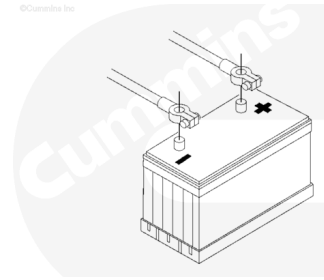
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Install the transmission. Refer to the original equipment manufacturer's instructions.

⚠ WARNING ⚠

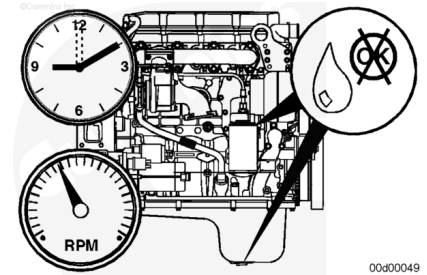
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Reconnect the batteries.

Operate the engine and check for leaks.



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